

A PROJECTIVE APPROACH TO NON-DIVIDING SETS

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ABSTRACT. Let $F(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ in which no element divides the sum of any nonempty subset of the remaining elements. We prove $F(N) = N^{1/5+o(1)}$, matching the constructions of Erdős. The proof is the density-increment argument of Pham and Zakharov for non-averaging sets, run one dimension lower: divisibility constraints survive the projective normalization $v \mapsto v/\lambda(v)$, which drops the associated convex geometry by one dimension and moves the exponent from $1/4$ to $1/5$. The proof has been formally verified in Lean 4. The novel idea was found by GPT 5.6 Sol.

1. INTRODUCTION

For a finite set $A \subseteq \mathbb{Z}_{>0}$, write

$$\Sigma(A) = \left\{ \sum_{a \in S} a : S \subseteq A \right\}$$

for its set of subset sums.

Definition 1.1. A finite set $A \subseteq \mathbb{Z}_{>0}$ is *non-dividing* if, for every $a \in A$ and every nonempty set $S \subseteq A \setminus \{a\}$,

$$a \nmid \sum_{s \in S} s.$$

Let $F(N)$ be the maximum cardinality of a non-dividing subset of $[N] = \{1, \dots, N\}$.

Erdős, Lev, Rauzy, Sándor and Sárközy introduced this problem and proved the bound $F(N) < 3N^{1/2} + 1$ [4]; it appears as Problem 131 in the collection of Erdős problems maintained by Bloom [1]. The previously known polynomial estimates were

$$N^{1/5} \ll F(N) \leq N^{1/4+o(1)}.$$

The lower bound is implicit in the constructions of Erdős and of Erdős–Lev–Rauzy–Sándor–Sárközy [3, 4]. The upper bound follows because every non-dividing set is non-averaging, together with the sharp theorem of Pham and Zakharov [5].

Our main result closes the gap between these exponents.

Theorem 1.2. *One has*

$$F(N) = N^{1/5+o(1)}.$$

Our proof is the density-increment argument of Pham and Zakharov [5], run one dimension lower. Every non-dividing set is non-averaging, since an averaging relation $\sum_{s \in S} s = |S|a$ is already forbidden modulo a ; this observation alone gives the previous upper bound, and since the exponent $1/4$ is best possible for non-averaging sets, any improvement must use the forbidden divisibilities themselves. The difficulty is that divisibility, unlike averaging, is destroyed by translation.

The one new idea is a projective normalization. A dyadic pigeonhole places a large part of the set in a shell $X \leq \lambda(v) \leq 2X$ of an integer linear functional λ (Section 2). The forbidden relations $\lambda(a) \mid \sum_{v \in S} \lambda(v)$ depend only on the values of λ , so they survive passage to subsets and

the coefficient identification of a proper homogeneous generalized arithmetic progression — the only operations we ever perform — and, crucially, they survive the normalization $\pi(v) = v/\lambda(v)$, which maps the shell into the affine hyperplane $\{\lambda = 1\}$ of dimension $d-1$. A tube-of-subset-sums argument (Theorem 5.4) shows that for an irreducible pair the projected family $\pi(V)$ must be in approximate convex position, and the convex-density increment of [5], applied in the hyperplane and lifted back to the lattice (Lemma 6.2), then runs with all convex geometry one dimension lower. This moves the exponents from the α_d of [5] to $\sigma_d = \alpha_{d-1}$ for $d \geq 2$, and to $\sigma_1 = \frac{1}{5}$ in the case $d = 1$ (Theorem 7.2), which is Theorem 1.2.

Apart from this modification, the derivation is that of [5]. Their structure theorem, progression-comparison lemmas, discrete John lemma, zonotope rounding lemma and convex-density lemma ([5, Corollary 5, Lemmas 6, 8, 7, 13 and 1]) are quoted and used as stated, while their preprocessing, slab and zonotope-filling steps and exponent numerology are needed in unshifted or projective variants, because translations are forbidden; for each such variant we state the form used and indicate the modifications to the proof in [5] rather than retranscribing it, and routine uniformity and bookkeeping verifications are likewise abbreviated. The complete proof, including every such adaptation, has been formally verified in Lean 4 over the mathlib library [8], with the quoted results of [5] and the Blaschke selection and volume-continuity theorems from [6] as the only non-foundational axioms.

2. THE PROJECTIVE FRAMEWORK

The upper-bound proof is most naturally formulated in boxes in $\mathbb{Z}^d$.

Definition 2.1 (Shell non-dividing set). Let $B \subseteq \mathbb{Z}^d$ be a symmetric axis-parallel box, let $\lambda : \mathbb{Z}^d \rightarrow \mathbb{Z}$ be a nonzero integer linear functional, and let $X > 0$. A finite set $V \subseteq B$ is (λ, X) -shell non-dividing if

$$X \leq \lambda(v) \leq 2X \quad (v \in V)$$

and, for every $a \in V$ and every nonempty $S \subseteq V \setminus \{a\}$,

$$\lambda(a) \nmid \sum_{v \in S} \lambda(v).$$

The shell restriction loses only a logarithmic factor in the original one-dimensional problem.

Definition 2.2 (Projective normalization). For v in the positive λ -shell, define

$$\pi(v) = \frac{v}{\lambda(v)}.$$

Then $\pi(v)$ belongs to the affine hyperplane

$$H_\lambda = \{y \in \mathbb{R}^d : \lambda(y) = 1\},$$

which has dimension $d-1$.

This is the source of the expected dimension drop.

2.1. The exponent sequence. Define

$$\sigma_1 = \frac{1}{5}, \quad \sigma_2 = \frac{1}{4}, \quad \sigma_d = \frac{d-2}{d} \quad (d \geq 3).$$

For $d \geq 2$, this equals the sharp Pham–Zakharov non-averaging exponent in dimension $d-1$.

3. STRUCTURAL INPUT AND UNSHIFTED PREPROCESSING

We use the higher-dimensional homogeneous subset-sum structure theorem of Conlon–Fox–Pham [2] in the form recorded by Pham–Zakharov [5, Corollary 5]. We state only the consequences required below. This section follows [5, Section 3] closely; the essential change throughout is that translations, which do not preserve divisibility, are never applied to the sets under consideration.

Throughout the technical argument, the ambient dimension and every displayed polynomial-box exponent are fixed. Parameters such as δ , γ and μ may depend on the set size n , and all $o(\cdot)$ relations and limits are taken as $n \rightarrow \infty$ along the family under consideration.

We use the following GAP conventions throughout. A symmetric homogeneous GAP is data

$$P = \left\{ \sum_{i=1}^e n_i q_i : |n_i| \leq N_i \right\}, \quad q_i \in \mathbb{Z}^\ell, \quad N_i \in \mathbb{Z}_{\geq 0}.$$

Its *dimension* is the number e of displayed steps, while its *rank* is $\dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{q_1, \dots, q_e\}$. It is *proper* if its coefficient map is injective on the displayed box. For $t \geq 0$ we set

$$tP = \left\{ \sum_{i=1}^e n_i q_i : |n_i| \leq \lfloor tN_i \rfloor \right\}.$$

If P is proper, ϕ_P denotes the inverse coefficient identification from P to its coordinate box. This is an identification on P , not in general an ambient homomorphism $\mathbb{Z}^\ell \rightarrow \mathbb{Z}^e$. For a real number $x \geq 3$ we also write

$$s(x) = \frac{x}{(\log x)^2};$$

thus $s(C) = s(|C|)$ below.

Theorem 3.1 (Conlon–Fox–Pham; Pham–Zakharov). *For every fixed ambient dimension ℓ and every $\beta > 1$ there are constants $c > 0$ and $D \in \mathbb{N}$ with the following property. Let $C \subseteq \mathbb{Z}^\ell$ have cardinality m and be contained in an axis-parallel box Q satisfying $|Q| \leq m^\beta$. For all sufficiently large m there are a set $\widehat{C} \subseteq C$ with*

$$|\widehat{C}| \geq m - c^{-1} \frac{m}{\log m},$$

a symmetric GAP

$$P(C) = \left\{ \sum_{i=1}^{d'} n_i q_i : |n_i| \leq N_i \right\}, \quad d' \leq D,$$

containing $\widehat{C} \cup \{0\}$, and a set $C_0 \subseteq \widehat{C}$ with $|C_0| \leq s(C)$ such that

$$q + cs(C)P(C) \subseteq \Sigma(C_0)$$

for some $q \in s(C)P(C)$. Moreover $cs(C)P(C)$ is proper.

The existence statement through properness is the cited corollary of Pham–Zakharov, derived from the Conlon–Fox–Pham structure theorem. Their symmetric convention is the centered display above, so $P(C)$ is homogeneous. The location $q \in s(C)P(C)$ is not an extra external hypothesis: since $C_0 \subseteq \widehat{C} \subseteq P(C)$ and $|C_0| \leq s(C)$, one has $\Sigma(C_0) \subseteq s(C)P(C)$; since $0 \in cs(C)P(C)$, the core inclusion itself gives $q \in \Sigma(C_0) \subseteq s(C)P(C)$.

For fixed ambient dimension and polynomial-box exponent, a *structural package* means a fixed choice of the constants and of the objects $\widehat{C}, C_0, q, P(C)$ in Theorem 3.1 for every eligible sufficiently large set C . When dimensions vary over a fixed finite range, one package is chosen in each ambient dimension before any iteration begins, and after every coefficient identification the package indexed by the new ambient dimension is used; all constants are then uniform once

the dimension range and a common polynomial-box exponent are fixed. Within each package we choose a structural GAP of minimum dimension and write $d(C) = \dim P(C)$; every radius then satisfies $N_i \geq 1$, for sufficiently large $|C|$ properness of $cs(C)P(C)$ implies properness of $P(C)$, and $\phi_{P(C)}$ is based at the origin. We suppress the chosen package from the notation.

Definition 3.2 (Unshifted irreducibility). Fix a structural package applicable to all sets under consideration. Let $V \subseteq B \subseteq \mathbb{Z}^d$, where B is a symmetric axis-parallel coordinate box. We say that the pair (V, B) is (δ, γ) -irreducible in the unshifted sense, relative to this package, if every $D \subseteq V$ with $|D| \geq \delta|V|$ has subset-sum dimension d and satisfies

$$|P(D)| \geq \gamma|B|.$$

This is [5, Definition 9] with the shifts $x \in \phi_P(P)$ of the witnessing subsets omitted.

The omission of translations is essential: shell non-divisibility is inherited by subsets and by coefficient identifications of proper homogeneous GAPs, with the shell functional transported as below, but not by arbitrary translations.

Lemma 3.3 (Transformation of the shell functional). *Let $P \subseteq \mathbb{Z}^\ell$ be a proper homogeneous GAP with step vectors $q_1, \dots, q_d$, and let $\lambda : \mathbb{Z}^\ell \rightarrow \mathbb{Z}$ be integer-linear. Define*

$$\lambda_P(n_1, \dots, n_d) = \sum_{i=1}^d n_i \lambda(q_i).$$

Then

$$\lambda_P(\phi_P(v)) = \lambda(v) \quad (v \in P).$$

Consequently, if $V \subseteq P$ is shell non-dividing for λ , then $\phi_P(V)$ is shell non-dividing for λ_P with exactly the same shell parameter.

Proof. Homogeneity gives $v = \sum_i n_i q_i$ whenever $\phi_P(v) = (n_1, \dots, n_d)$. The identity follows by linearity. Every prohibited divisibility relation is an identity involving only the integer values of the functional, so it is preserved exactly. $\square$

We will use the progression-comparison lemmas [5, Lemmas 6 and 8] as stated: every box arising below — the ambient box B and the coefficient boxes of the structural GAPs — is a symmetric axis-parallel coordinate box, so both lemmas apply literally, with constants uniform over bounded ranges of the ambient and structural dimensions.

Lemma 3.4 (Unshifted preprocessing). *Fix an ambient dimension ℓ , $\beta > 1$, $\varepsilon \in (0, 1/3)$ and a desired shrink exponent $K_0 \geq 1$. Let $V \subseteq B \subseteq \mathbb{Z}^\ell$ be shell non-dividing, with $|B| \leq |V|^\beta$. Suppose $0 < \delta \leq 1/2$ and*

$$\gamma \leq \delta^K, \quad \gamma \geq |V|^{-1/3},$$

where $K = K(\ell, \beta, \varepsilon, K_0)$ is sufficiently large. Then one can pass, using only subsets and coefficient identifications of proper homogeneous GAPs, to a shell non-dividing set $\tilde{V} \subseteq \tilde{B} \subseteq \mathbb{Z}^{\tilde{d}}$ and fix a structural package in $\mathbb{Z}^{\tilde{d}}$ such that $(\tilde{V}, \tilde{B})$ is (δ, γ) -irreducible relative to that package and

$$\begin{aligned} |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} |V|^{-(1-\varepsilon)(\tilde{d}-\ell)} |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &> \ell, \\ |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} \rho^{K_0} |B|, & |\tilde{V}| &= \rho |V| > |V|^{1-\varepsilon}, & \tilde{d} &= \ell, \\ |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &< \ell. \end{aligned}$$

Here $\tilde{d} = O_{\ell, \beta}(1)$; this dimension bound is uniform for $\varepsilon \in (0, 1/3)$ and $K_0 \geq 1$. The polynomial-box exponent and constants of the output structural package are bounded in $\ell, \beta, \varepsilon, K_0$.

Proof. This is the unshifted analogue of [5, Lemma 10]. We run the same iteration of down-, up- and shrink-moves and indicate the three modifications; the accompanying bookkeeping, abbreviated here, is carried out in full in the formalization [8].

Put $n = |V|$ and $\delta_* = \delta/2$, and assume throughout that n is sufficiently large in the fixed parameters. Before the iteration begins, fix the initial structural package for (ℓ, β) , constants

$$C_U > 3(\beta + 2), \quad \bar{d} > D(\ell, \beta) + C_U, \quad \bar{\beta} = 4(\beta + 3),$$

structural packages with common polynomial-box exponent $\bar{\beta}$ in every ambient dimension $1 \leq r \leq \bar{d}$, and a constant $C = C(\bar{d}, \bar{\beta})$ dominating the constants of [5, Lemmas 6 and 8] for ambient dimensions at most $\bar{d}$ and structural dimensions at most $\max_{r \leq \bar{d}} D(r, \bar{\beta})$. At an accepted stage A_j (with $A_0 = V$), write $P_j = P(A_j)$, $d_j = \dim P_j$, $A_j^\sharp = \phi_{P_j}(\hat{A}_j)$ and $Q_j = \phi_{P_j}(P_j)$, and stop before accepting a move whose cardinality is at most $n^{1-\varepsilon/2}$ or whose structural dimension exceeds $\bar{d}$.

Translations are omitted. A move of [5, Lemma 10] passes from the current set to $A_1 - x$ for a witness $A_1 \subseteq A_j^\sharp$ with $|A_1| \geq \delta|A_j^\sharp|$ and a shift $x \in Q_j$. Here witnesses are taken with $x = 0$ only, matching Definition 3.2. This restricts the available moves and changes no estimate: the down-, up- and shrink-move bounds

$$|P_{j+1}| \leq C|P_j|, \quad |P_{j+1}| \leq Cs(A_{j+1})^{-(d_{j+1}-d_j)}|P_j|, \quad |P_{j+1}| < \gamma|P_j|$$

are supplied by [5, Lemmas 6 and 8], applied in the coordinate box Q_j , which also give the initial bound $|P_0| \ll_{\ell, \beta} n^{-(1-\varepsilon)\max(0, d_0-\ell)}|B| \leq n^{\beta+1}$.

The shell functional is transported. At each coefficient identification the functional is replaced by λ_{P_j} as in Lemma 3.3, so every accepted set is shell non-dividing with the same shell parameter. This replaces the (translation-invariant) preservation of the non-averaging property used in [5], and is the reason translations must be omitted.

Packages are fixed in advance. Every application of the structure theorem after the initial one uses the pre-selected package in the current ambient dimension d_j . This is legitimate because the eligibility condition $|Q_j| \leq (\delta|A_j^\sharp|)^{\bar{\beta}}$ holds at every accepted stage: discarding all factors at most one in the product of the move estimates gives $|P_j| \leq C^{D(\ell, \beta)+2C_U}|P_0| \leq n^{\beta+2}$, while $\gamma \geq n^{-1/3}$, $K \geq 1$ and $|A_j| > n^{1-\varepsilon/2}$ give $\delta|A_j^\sharp| \geq \frac{1}{2}n^{2/3-\varepsilon/2}$, whose $\bar{\beta}$ -th power exceeds $n^{\beta+2}$ because $\varepsilon < 1/3$ and $\bar{\beta} = 4(\beta + 3)$. Hence no post hoc choice of structural constants is ever made.

With these modifications the analysis of [5, proof of Lemma 10] applies verbatim. Let $A_0, \dots, A_i$ be the accepted stages. Multiplying the move estimates and using $|P_i| \geq 1$ bounds the total upward dimension change by $U \leq \frac{2(\beta+2)}{1-\varepsilon} \leq 3(\beta + 2) < C_U$, so the dimension cutoff never triggers. A proposed set of cardinality at most $n^{1-\varepsilon/2}$ would satisfy $\delta_*^{i+1} \leq n^{-\varepsilon/2}$; since $\gamma \leq \delta^K \leq \delta_*^{K/2}$ and $|P_0| \leq n^{\beta+1}$, the product bound then forces $|P_i| < 1$ once K is sufficiently large in $\ell, \beta, \varepsilon, K_0$, so the cardinality cutoff never triggers either. The procedure therefore stops only because no move is available, and the terminal pair

$$\tilde{V} = A_i^\sharp, \quad \tilde{B} = Q_i, \quad \tilde{d} = d_i$$

is (δ, γ) -irreducible in the unshifted sense relative to the fixed package in dimension d_i , with $|\tilde{V}| > |V|^{1-\varepsilon}$ by the deletion estimate; the terminal package belongs to the finite preselected collection, whose data depend only on $\ell, \beta, \varepsilon, K_0$.

The three box estimates follow from the product bound exactly as at the end of [5, proof of Lemma 10]. Writing $\mathcal{D}, \mathcal{U}$ for the sets of down- and up-move indices and $M = U + \max(0, d_0 - \ell)$, the product bound reads $|P_i| \ll_{\ell, \beta, \varepsilon} \gamma^{i-|\mathcal{U}|-|\mathcal{D}|} n^{-(1-\varepsilon)M} |B|$. If $\tilde{d} > \ell$, then $M \geq \tilde{d} - \ell$, giving the

first alternative; if $\tilde{d} < \ell$, discarding factors at most one gives the third. If $\tilde{d} = \ell$, the dimension bookkeeping gives $|\mathcal{U}| + |\mathcal{D}| \leq 2M$, and $\varepsilon < 1/3$ with $\gamma \geq n^{-1/3}$ gives $n^{-(1-\varepsilon)M} \leq \gamma^{2M}$, whence $|P_i| \ll \gamma^i |B|$; since $\rho = |\tilde{V}|/n \gg \delta_*^i$ and $\gamma \leq \delta_*^{K/2}$, choosing $K \geq 2K_0$ sufficiently large gives $\gamma^i \ll_{K_0} \rho^{K_0}$, the same-dimensional alternative. $\square$

4. THREE GEOMETRIC-ARITHMETIC LEMMAS

We next establish the slab, rounding and lattice-index statements used in the tube argument. Most of them are unshifted or homogeneous variants of estimates from [5] and are proved by reference; the index and span lemmas at the end of the section (Lemmas 4.5 and 4.6) are new.

Throughout this section and the next, $B_{\mathbb{R}} = \prod_{i=1}^d [-M_i, M_i]$ with $M_i \in \mathbb{Z}_{\geq 1}$ is a symmetric real box, $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$, and $w_B(u) = \max_{x \in B_{\mathbb{R}}} |u(x)|$ for a linear functional u . The following hypotheses are common to several statements below, and we refer to them jointly as the *irreducible setting*: d and $\beta > 1$ are fixed, together with a structural package arising from Theorem 3.1 in ambient dimension d with polynomial-box exponent at most β , applicable to every subset of V of size at least $\delta|V|$; the pair (V, B) , where $V \subseteq B \subseteq \mathbb{Z}^d$, is unshiftedly (δ, γ) -irreducible relative to this package; and $n = |V|$ satisfies $|B| \leq n^\beta$.

4.1. A slab dichotomy.

Lemma 4.1 (Convex lattice sets and their John progression). *Let $H \subseteq \mathbb{R}^d$ be an r -dimensional rational subspace, let $\Lambda = H \cap \mathbb{Z}^d$, and let $K \subseteq H$ be an origin-symmetric convex body such that $K \cap \Lambda$ spans H . Then $K \cap \Lambda$ is contained in a proper symmetric rank- r GAP $Q \subseteq \Lambda$ satisfying*

$$|Q| \ll_d |K \cap \Lambda|.$$

If $r = d$, then additionally

$$|K \cap \mathbb{Z}^d| \ll_d \text{vol}_d(K).$$

Proof. The case $r = 0$ is immediate. For $r \geq 1$, identify Λ with $\mathbb{Z}^r$; the containment of $K \cap \Lambda$ in a proper symmetric rank- r GAP Q with $|Q| \ll_r |K \cap \Lambda|$ is exactly the discrete John lemma [5, Lemma 7]. If $r = d$, the successive-minima point count and Minkowski's second theorem (see [7, Chapter 3]) give $|K \cap \mathbb{Z}^d| \ll_d \prod_{i=1}^d \lambda_i^{-1} \ll_d \text{vol}_d(K)$, the last successive minimum being at most one because $K \cap \mathbb{Z}^d$ spans $\mathbb{R}^d$. $\square$

Lemma 4.2 (Slab exclusion from irreducibility). *Assume the irreducible setting, and suppose*

$$\delta \geq n^{-1/2}, \quad [s(\delta n)]^{-1} = o(\gamma).$$

There is $c_{d,\beta} > 0$ such that, for every nonzero linear functional u and every $0 < \tau \leq c_{d,\beta}\gamma$, the central slab $\{x : |u(x)| \leq \tau w_B(u)\}$ contains fewer than δn points of V .

Proof. This is the unshifted form of the slab claim in the proof of [5, Lemma 14]. Suppose that $D \subseteq V$ lies in the slab and $|D| \geq \delta n$. Since $\delta \geq n^{-1/2}$, one has $|B| \leq n^\beta \leq |D|^{2\beta}$, so the fixed structural package is applicable to D , and irreducibility gives $d(D) = d$ and $|P(D)| \geq \gamma|B|$.

Let $K = B_{\mathbb{R}} \cap \{x : |u(x)| \leq \tau w_B(u)\}$ and let r be the dimension of $H = \text{span}_{\mathbb{R}}(K \cap \mathbb{Z}^d)$. Lemma 4.1, applied to the body $K \cap H$ in the lattice $H \cap \mathbb{Z}^d$, places $K \cap \mathbb{Z}^d \supseteq D$ in a proper symmetric rank- r GAP Q with $|Q| \ll_d |K \cap \mathbb{Z}^d|$. If $r < d$, then trivially $|Q| \ll_d |B|$. If $r = d$, then moreover $|K \cap \mathbb{Z}^d| \ll_d \text{vol}_d(K) \leq d\tau \text{vol}_d(B_{\mathbb{R}}) \asymp_d \tau|B|$, because, writing $u(x) = \sum_i a_i x_i$ and choosing j with $|a_j| M_j \geq w_B(u)/d$, the x_j -sections of K are intervals of length at most $2\tau w_B(u)/|a_j| \leq 2d\tau M_j$.

Put $s_D = s(D) \geq s(\delta n)$. Comparing the proper GAP $q + cs_D P(D) \subseteq \Sigma(D_0) \subseteq s_D Q$ gives $s_D^d |P(D)| \ll_{d,\beta} s_D^r |Q|$. If $r < d$, this yields $|P(D)| \ll_{d,\beta} s_D^{-1} |B| = o(\gamma|B|)$ by the parameter

hypothesis; if $r = d$, it yields $|P(D)| \ll_{d,\beta} \tau|B| \leq C_{d,\beta} c_{d,\beta} \gamma|B|$. Choosing $c_{d,\beta}$ sufficiently small contradicts $|P(D)| \geq \gamma|B|$ in either case. $\square$

4.2. One-sided filling around a zonotope. For a finite set $E \subseteq \mathbb{R}^d$, write

$$Z(E) = \left\{ \sum_{v \in E} t_v v : 0 \leq t_v \leq 1 \right\}$$

for its zonotope. For a homogeneous GAP P , let $\langle P \rangle$ be the lattice generated by its steps.

Lemma 4.3 (Rounding with a homogeneous core). *Let $C \subseteq B \subseteq \mathbb{Z}^d$, let $P = P(C)$ have dimension d , and assume that its step lattice*

$$L = \langle P \rangle$$

has rank d . Take the objects $\hat{C}, C_0, q$ from Theorem 3.1, and put

$$\overline{C} = \hat{C} \setminus C_0, \quad s = s(C).$$

For all sufficiently large $|C|$,

$$L \cap \left(Z(\overline{C}) + q + \frac{cs}{2}P \right) \subseteq \Sigma(C).$$

Proof. This inclusion is established, from the rounding lemma [5, Lemma 13], in the proof of [5, Theorem 4]; we indicate the form the derivation takes in the homogeneous full-rank setting. Since P has exactly d steps and L has rank d , the steps are linearly independent, hence a $\mathbb{Z}$ -basis of L , and the coefficient map extends to a real-linear isomorphism ψ with $\psi(P) = \prod_{i=1}^d [-N_i, N_i] \cap \mathbb{Z}^d$. Let $y = z + t$ with $z \in Z(\overline{C})$ and $t \in q + (cs/2)P$. Since $q \in sP \subseteq L$, also $z = y - t \in L$. Applying [5, Lemma 13] to $\psi(\overline{C}) \cup \{0\}$ in the coefficient box yields $\tilde{z} \in \Sigma(\overline{C})$ with $z - \tilde{z} \in 3\sqrt{2d|C|}P$, and $3\sqrt{2d|C|} \leq cs/2$ for all sufficiently large $|C|$ because $s = |C|/(\log |C|)^2$. Coefficientwise addition in the symmetric GAP then gives $(z - \tilde{z}) + t \in q + csP \subseteq \Sigma(C_0)$, and since $\overline{C}$ and C_0 are disjoint, $y = \tilde{z} + ((z - \tilde{z}) + t) \in \Sigma(C)$. $\square$

4.3. A large progression has small lattice index.

Lemma 4.4 (Containment of the structural GAP). *Let $C \subseteq B \subseteq \mathbb{Z}^d$, where $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$ and $B_{\mathbb{R}}$ is a symmetric axis-parallel real box. Let $P = P(C)$ and let c be the constant in Theorem 3.1. For all sufficiently large $|C|$,*

$$P \subseteq \frac{4}{c}B_{\mathbb{R}}.$$

In particular, when the ambient dimension and polynomial-box exponent are fixed, $P \subseteq C_{d,\beta}B_{\mathbb{R}}$.

Proof. This is the homogeneous form of the containment step in [5, Lemma 11]. Let $s = s(C)$ and let C_0, q be the structural core and translation. Since $C_0 \subseteq B$ and $|C_0| \leq s$,

$$q + csP \subseteq \Sigma(C_0) \subseteq sB_{\mathbb{R}}.$$

As $0 \in csP$, also $q \in sB_{\mathbb{R}}$, and hence $csP \subseteq 2sB_{\mathbb{R}}$. For $p \in P$, put $k = \lfloor cs \rfloor \geq cs/2$. Then $kp \in csP$, so $p \in (2s/k)B_{\mathbb{R}} \subseteq (4/c)B_{\mathbb{R}}$. $\square$

Lemma 4.5 (Full rank and index). *Assume the irreducible setting. Let $C \subseteq V$ have $|C| \geq \delta|V|$ and assume $|B| \leq |C|^\beta$. Put*

$$P = P(C), \quad s = s(C), \quad L = \langle P \rangle.$$

If

$$s\gamma \longrightarrow \infty,$$

then L has rank d and

$$[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}.$$

Proof. Irreducibility gives $\dim P = d$ and $|P| \geq \gamma|B|$. Let $c = c(d, \beta) > 0$ be the structural constant in Theorem 3.1; the proof of Lemma 4.4 gives the stronger core containment

$$csP \subseteq 2sB_{\mathbb{R}}. \quad (1)$$

Write $P = \{\sum_{i=1}^d n_i q_i : |n_i| \leq N_i\}$, where every $N_i \geq 1$ as zero-width directions are not counted in the dimension, and let $H = \text{span}_{\mathbb{R}}\{q_1, \dots, q_d\}$, of dimension r . Because csP is proper, its cardinality is the product of its coefficient widths, so for all sufficiently large s

$$|csP| = \prod_{i=1}^d (2\lfloor csN_i \rfloor + 1) \gg_{d,\beta} s^d |P|, \quad (2)$$

while (1) gives

$$|csP| \leq |H \cap 2sB_{\mathbb{R}} \cap \mathbb{Z}^d| \leq (2s)^r |B|, \quad (3)$$

the second inequality by projecting to r coordinates on which H injects. Combining (2) and (3) with $|P| \geq \gamma|B|$ yields $s^{d-r}\gamma \ll_{d,\beta} 1$. If $r < d$, the left-hand side is at least $s\gamma$ for large s , contradicting the hypothesis. Hence $r = d$, the d step vectors are linearly independent, and they form a $\mathbb{Z}$ -basis of their integer span L .

Let M be the $d \times d$ matrix whose columns are $q_1, \dots, q_d$, so that $L = M\mathbb{Z}^d$ and $[\mathbb{Z}^d : L] = |\det M|$. Since $\text{conv } P = M(\prod_{i=1}^d [-N_i, N_i])$,

$$\text{vol}_d(\text{conv } P) = 2^d |\det M| \prod_{i=1}^d N_i \asymp_d [\mathbb{Z}^d : L] |P|, \quad (4)$$

the comparison using $N_i \geq 1$ and therefore uniform in the widths. Lemma 4.4 and convexity give $\text{conv } P \subseteq C_{d,\beta} B_{\mathbb{R}}$, and $\text{vol}_d(B_{\mathbb{R}}) \asymp_d |B|$, so equation (4) implies $[\mathbb{Z}^d : L] |P| \ll_{d,\beta} |B|$. Using $|P| \geq \gamma|B|$ now gives $[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}$, as claimed. $\square$

Lemma 4.6 (Full-dimensional projective support). *Assume the irreducible setting, with $V(\lambda, X)$ -shell non-dividing, and suppose $[s(\delta n)]^{-1} = o(\gamma)$. Then, for all sufficiently large n ,*

$$\text{span}_{\mathbb{R}} V = \mathbb{R}^d, \quad \text{aff } \pi(V) = H_{\lambda}.$$

Moreover

$$\pi(V) \subseteq \Omega := \{y \in H_{\lambda} : Xy \in B_{\mathbb{R}}\},$$

and Ω has nonempty relative interior in H_{λ} .

Proof. Apply Lemma 4.5 with $C = V$. Since $s(V) \geq s(\delta n)$ for all sufficiently large n , the hypothesis implies $s(V)\gamma \rightarrow \infty$. The structural step lattice $L = \langle P(V) \rangle$ therefore has rank d .

Let $V_0 \subseteq V$ and q be the structural core and translation from Theorem 3.1. For sufficiently large n , every coordinate radius of $cs(V)P(V)$ is at least one. Since

$$q + cs(V)P(V) \subseteq \Sigma(V_0) \subseteq \text{span}_{\mathbb{R}} V,$$

taking differences of points in the left-hand translate shows that every step of $P(V)$ belongs to $\text{span}_{\mathbb{R}} V$. These steps have rank d , so $\text{span}_{\mathbb{R}} V = \mathbb{R}^d$.

If $\pi(V)$ were contained in a proper affine subspace of H_{λ} , then its linear span would have dimension at most $d-1$. Since $v = \lambda(v)\pi(v)$ for every $v \in V$, this would put V in a proper linear subspace of $\mathbb{R}^d$, a contradiction. Hence $\text{aff } \pi(V) = H_{\lambda}$.

Finally, for $v \in V$ the shell condition gives $X/\lambda(v) \in [1/2, 1]$. As $B_{\mathbb{R}}$ is convex, symmetric and contains the origin,

$$X\pi(v) = \frac{X}{\lambda(v)}v \in B_{\mathbb{R}}.$$

Thus $\pi(V) \subseteq \Omega$. Since Ω is convex and contains an affinely spanning subset of H_{λ} , it has nonempty relative interior. $\square$

5. PROJECTIVE CONVEXITY FROM A TUBE OF SUBSET SUMS

We now prove the central projective statement.

Definition 5.1 (Convex position for an indexed multiset). Let E be a finite-dimensional Euclidean affine space. An indexed family $p_1, \dots, p_n \in E$ is in μ -convex position if, for every index j , there is a closed half-space of E containing p_j and containing at most μn indices, counted with multiplicity. In a zero-dimensional affine space the only nonempty half-space is the whole space. This is the indexed variant of the δ -convex position of [5, Section 2].

Lemma 5.2 (Projective duality). *Let V be a finite set lying in a (λ, X) -shell, and suppose that the indexed projective family $\pi(V)$ is not in μ -convex position. If $\mu n \geq 4$, then there are $a \in V$ and coefficients β_v for $v \in V \setminus \{a\}$ such that*

$$a = \sum_{v \neq a} \beta_v v, \quad 0 \leq \beta_v \leq \frac{4}{\mu n}, \quad \frac{1}{3} \leq \sum_{v \neq a} \beta_v \leq 3.$$

Proof. Choose an index a for which the convex-position condition fails. The separation argument at the start of the proof of [5, Theorem 4], applied to the projected family, yields coefficients $0 \leq c_v \leq (\mu n)^{-1}$ with $\sum_v c_v = 1$ and

$$\pi(a) = \sum_v c_v \pi(v) :$$

if no such coefficients existed, a functional separating 0 from the compact convex set of all combinations $\sum_v c_v(\pi(v) - \pi(a))$ would be negative on that set, while failure of convex position places more than μn indices in the closed half-space where the functional is nonnegative, and distributing the unit mass over those indices gives a contradiction. Multiplying by $\lambda(a)$ gives

$$a = \sum_v c_v \frac{\lambda(a)}{\lambda(v)} v.$$

The shell condition puts every ratio $\lambda(a)/\lambda(v)$ in $[1/2, 2]$. Move the possible $v = a$ term to the left and divide by $1 - c_a$. Since $c_a \leq (\mu n)^{-1} \leq 1/4$, the stated bounds follow. $\square$

Lemma 5.3 (Robust weighted zonotope). *Assume the irreducible setting, and suppose*

$$\delta \geq n^{-1/2}, \quad \delta = o(\mu), \quad [s(\delta n)]^{-1} = o(\gamma).$$

Let $a \in V$ and let weights $(\theta_v)_{v \in V \setminus \{a\}}$ satisfy

$$0 \leq \theta_v \leq \frac{1}{2}, \quad c_1 \mu n \leq \sum_{v \neq a} \theta_v \leq c_2 \mu n$$

for fixed positive constants c_1, c_2 . Let $R \subseteq V \setminus \{a\}$ have size $o(\mu n)$, put $C = (V \setminus \{a\}) \setminus R$, and set

$$z_C = \sum_{v \in C} \theta_v v.$$

Then, for a constant $c_{d,\beta,c_1,c_2} > 0$,

$$z_C + c_{d,\beta,c_1,c_2} \mu \gamma n B_{\mathbb{R}} \subseteq Z(C).$$

Proof. This is [5, Lemma 14] in the unshifted weighted setting, and we follow its proof. Let $Z^0 = \sum_{v \in C} [-\theta_v, \theta_v]v$; it suffices to prove $\xi n B_{\mathbb{R}} \subseteq Z^0$ with $\xi = c_{d,\beta,c_1,c_2} \mu \gamma$. If $y \in \xi n B_{\mathbb{R}} \setminus Z^0$, separation gives a linear functional u with $u(y) > \sum_{v \in C} \theta_v |u(v)|$. Put $z = u(y) > 0$ and $H = \{x : |u(x)| < c'_d \gamma z / (\xi n)\}$, a central slab of relative width at most $c'_d \gamma$ because $y \in \xi n B_{\mathbb{R}}$. In place of the slab claim of [5], which invokes shifted irreducibility, we apply Lemma 4.2: $|C \cap H| < \delta n$. The mass count is then identical to that in [5]: the total weight is $\asymp \mu n$, removing R costs $o(\mu n)$, and the weight on H is at most $\delta n/2 = o(\mu n)$, so $\sum_{v \in C \setminus H} \theta_v \gg \mu n$; since $|u(v)| \geq c'_d \gamma z / (\xi n)$ off H , this gives $\sum_{v \in C} \theta_v |u(v)| \gg \mu n \cdot \gamma z / (\xi n) > z$ once c_{d,β,c_1,c_2} is sufficiently small, contradicting the separating inequality. $\square$

Theorem 5.4 (Projective convexity). *Assume the irreducible setting, with V (λ, X) -shell non-dividing. Assume $0 < \delta, \gamma, \mu < 1$ and the parameter hierarchy*

$$\delta \geq n^{-1/2}, \quad \delta = o(\mu), \quad \frac{n}{\log n} = o(\mu \gamma n), \quad \gamma^{-1} = o(\mu \gamma n), \quad [s(\delta n)]^{-1} = o(\gamma).$$

Then the indexed family $\pi(V)$ is in μ -convex position for all sufficiently large n .

Proof. Suppose otherwise, and use Lemma 5.2 to obtain a and coefficients β_v . Choose a sufficiently small absolute constant $c_0 > 0$ and set

$$T = \lfloor c_0 \mu n \rfloor, \quad \theta_v = T \beta_v.$$

Then

$$Ta = \sum_{v \neq a} \theta_v v, \quad 0 \leq \theta_v \leq \frac{1}{2}, \quad \sum_{v \neq a} \theta_v \asymp \mu n.$$

Put $C = V \setminus \{a\}$ and let $\widehat{C}, C_0, P, q$ be the objects assigned to C by the fixed structural package. Put

$$L = \langle P \rangle, \quad \overline{C} = \widehat{C} \setminus C_0,$$

and write $s = s(C)$. For sufficiently large n , one has $|C| \geq \delta n$. Irreducibility therefore gives $d(C) = d$ and $|P| \geq \gamma |B|$. Moreover, the hypothesis $[s(\delta n)]^{-1} = o(\gamma)$ implies $s(C) \gamma \rightarrow \infty$. Since $|C| = n - 1$, the bound $|B| \leq n^\beta$ implies $|B| \leq |C|^{2\beta}$ for sufficiently large n . Lemma 4.5, with polynomial-box exponent 2β , consequently applies before any rounding is performed and gives

$$\text{rank } L = d, \quad [\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}.$$

The discarded set $R = C \setminus \overline{C}$ has size $O(n/\log n) = o(\mu n)$, so Lemma 5.3 gives

$$\bar{z} + \xi_0 n B_{\mathbb{R}} \subseteq Z(\overline{C}), \quad \bar{z} = \sum_{v \in \overline{C}} \theta_v v, \quad \xi_0 \asymp_{d,\beta} \mu \gamma,$$

while $Ta - \bar{z} \in O(n/\log n) B_{\mathbb{R}}$. Lemma 4.4 gives $P \subseteq C_{d,\beta} B_{\mathbb{R}}$, and $q \in sP$; hence

$$(Ta - \bar{z}) - q \in O\left(\frac{n}{\log n} + s\right) B_{\mathbb{R}} = o(\xi_0 n) B_{\mathbb{R}},$$

so the choice $\xi = \xi_0/2$ gives, for all sufficiently large n ,

$$(Ta - \bar{z}) - q + \xi n B_{\mathbb{R}} \subseteq \xi_0 n B_{\mathbb{R}}.$$

Consequently, if $y \in L \cap (Ta + \xi n B_{\mathbb{R}})$, then $y - q - \bar{z} \in \xi_0 n B_{\mathbb{R}}$, so $y - q \in Z(\overline{C})$ and $y \in Z(\overline{C}) + q + \frac{cs}{2} P$; Lemma 4.3 therefore gives the tube inclusion

$$\boxed{L \cap (Ta + \xi n B_{\mathbb{R}}) \subseteq \Sigma(C)}. \quad (\dagger)$$

Let r be the order of $a + L$ in $\mathbb{Z}^d/L$. Then $r \ll_{d,\beta} \gamma^{-1}$ and $ra \in L$. Since $r = o(\xi n)$ and $r = o(T)$, the nearest multiple of r to T is positive; call it m_0 . It satisfies

$$|m_0 - T| \leq \frac{r}{2} < \xi n.$$

As $a \in B$,

$$m_0 a \in L \cap (Ta + \xi n B_{\mathbb{R}}).$$

The tube inclusion gives $m_0 a \in \Sigma(V \setminus \{a\})$. Applying λ yields

$$m_0 \lambda(a) = \sum_{v \in S} \lambda(v)$$

for a nonempty $S \subseteq V \setminus \{a\}$, contradicting shell non-divisibility. $\square$

6. WEIGHTED PROJECTIVE DENSITY INCREMENT

We require an indexed version of the convex-density lemma. Rather than retranscribing the geometric proof, we derive it from the published set version by a perturbation and compactness argument.

Lemma 6.1 (Weighted convex-density lemma). *Fix $r \geq 1$ and $\varepsilon > 0$. There are $\tau \in (0, 1)$ and $\mu_0 > 0$ such that the following holds. Let $p_1, \dots, p_n$ be an indexed family in a convex body $\Omega \subseteq \mathbb{R}^r$ with nonempty interior, in μ -convex position, where $0 < \mu < \mu_0$, and assume that n is sufficiently large in r, ε and μ . Then there are*

$$\eta \in [\mu, \mu^\tau]$$

and a compact convex set $\Omega' \subseteq \Omega$ such that

$$\text{vol}_r(\Omega') \leq \eta \text{vol}_r(\Omega)$$

and

$$\#\{i : p_i \in \Omega'\} \geq \eta^{\kappa_r + \varepsilon} n, \quad \kappa_r = \frac{r-1}{r+1}.$$

For $r = 1$ this means $\kappa_1 = 0$.

Proof. Take τ and μ_0 from [5, Lemma 1]. For each j , choose a witnessing closed half-space H_j containing p_j and at most μn indices. For $m \geq 1$, perturb the points successively to distinct points $q_i^{(m)} \in H_i \cap B(p_i, 1/m)$, chosen so close to p_i that $q_i^{(m)} \in H_j$ only when $p_i \in H_j$; this is possible because only finitely many half-spaces H_j omit p_i , each at positive distance from it, and $H_i \cap B(p_i, 1/m)$ is infinite as $r \geq 1$. The set $Q_m = \{q_i^{(m)}\}$ is then in μ -convex position inside $\Omega_m = \Omega + m^{-1} \overline{B_2^r}$, so [5, Lemma 1] gives $\eta_m \in [\mu, \mu^\tau]$ and a compact convex $K_m \subseteq \Omega_m$ with $\text{vol}_r(K_m) \leq \eta_m \text{vol}_r(\Omega_m)$ and $|Q_m \cap K_m| \geq \eta_m^{\kappa_r + \varepsilon} n$. Pass to a subsequence on which $\eta_m \rightarrow \eta$, the finite index set $\{i : q_i^{(m)} \in K_m\}$ is constant, and K_m converges in the Hausdorff metric to a compact convex set K (Blaschke selection; convex-body volume is continuous under Hausdorff convergence [6]). Since Ω_m decreases to Ω , one has $K \subseteq \Omega$, every stabilized index yields $p_i \in K$, and $\Omega' = K$ has the required properties. $\square$

We now lift a projective cap back to the lattice.

Lemma 6.2 (Projective cap lifting). *Let $B_{\mathbb{R}} \subseteq \mathbb{R}^d$ be a symmetric coordinate box, let $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$, and let $\lambda : \mathbb{R}^d \rightarrow \mathbb{R}$ be a nonzero integer linear functional. Equip*

$$H_\lambda = \{y \in \mathbb{R}^d : \lambda(y) = 1\}$$

with its induced Euclidean $(d-1)$ -dimensional measure m_λ , and put

$$\Omega = \{y \in H_\lambda : Xy \in B_{\mathbb{R}}\}.$$

Let $\Omega' \subseteq \Omega$ be compact and convex with

$$m_\lambda(\Omega') \leq \eta m_\lambda(\Omega).$$

Finally, let

$$W = \{v \in B : X \leq \lambda(v) \leq 2X, \pi(v) \in \Omega'\}.$$

Then there are a symmetric coordinate box $B' \subseteq \mathbb{Z}^r$, an integer-linear functional $\lambda' : \mathbb{Z}^r \rightarrow \mathbb{Z}$, and an injective map $\phi : W \rightarrow B'$ such that

$$\lambda'(\phi(v)) = \lambda(v) \quad (v \in W),$$

and one of the following holds:

- (i) $r < d$ and $|B'| \ll_d |B|$;
- (ii) $r = d$ and $|B'| \ll_d \eta |B|$.

In particular, shell non-divisibility and the shell parameter are preserved exactly under the identification.

Proof. Write $\lambda(x) = \ell \cdot x$ for a nonzero vector $\ell \in \mathbb{Z}^d$. For a measurable set $E \subseteq H_\lambda$ and $T > 0$, put $\text{Cone}_T(E) = \{ty : y \in E, 0 \leq t \leq T\}$. Integrating the Jacobian $t^{d-1}/\|\ell\|_2$ on $H_\lambda = \ell/\|\ell\|_2^2 + \ker \lambda$ gives

$$\text{vol}_d(\text{Cone}_T(E)) = \frac{T^d}{d\|\ell\|_2} m_\lambda(E), \quad (5)$$

so the factor depending on λ cancels from volume ratios.

Define $K = B_\mathbb{R} \cap \text{Cone}_{2X}(\Omega')$; it is convex, contains 0, and contains W . Since $\text{Cone}_X(\Omega) \subseteq B_\mathbb{R}$ (as $Xy \in B_\mathbb{R}$ for $y \in \Omega$ and $B_\mathbb{R}$ is convex and contains the origin), formula (5) gives

$$\begin{aligned} \text{vol}_d(K) &\leq \text{vol}_d(\text{Cone}_{2X}(\Omega')) \\ &\leq 2^d \eta \text{vol}_d(\text{Cone}_X(\Omega)) \leq 2^d \eta \text{vol}_d(B_\mathbb{R}). \end{aligned} \quad (6)$$

Put $S = (K - K) \cap \mathbb{Z}^d$, $H = \text{span}_\mathbb{R} S$ and $r = \dim H$. Since $0 \in K$ and $W \subseteq K \cap \mathbb{Z}^d$, one has $W \subseteq S$; also $K - K \subseteq 2B_\mathbb{R}$. Applying Lemma 4.1 in the lattice $H \cap \mathbb{Z}^d$ to the symmetric convex body $(K - K) \cap H$ gives a proper symmetric homogeneous rank- r GAP $Q \subseteq H \cap \mathbb{Z}^d$ such that

$$S \subseteq Q, \quad |Q| \ll_d |S|. \quad (*)$$

If $r < d$, then, writing $B_\mathbb{R} = \prod_i [-M_i, M_i]$,

$$|S| \leq |2B_\mathbb{R} \cap \mathbb{Z}^d| = \prod_{i=1}^d (4M_i + 1) \leq 2^d |B|,$$

so (*) gives $|Q| \ll_d |B|$. If $r = d$, then $K - K$, and hence K , is full-dimensional, and the full-rank part of Lemma 4.1, the Rogers–Shephard inequality [6] and (6) give

$$|Q| \ll_d |S| \ll_d \text{vol}_d(K - K) \leq \binom{2d}{d} \text{vol}_d(K) \ll_d \eta \text{vol}_d(B_\mathbb{R}) \ll_d \eta |B|.$$

Finally write $Q = \{\sum_{i=1}^r n_i q_i : |n_i| \leq N_i\}$ and put

$$B' = \prod_{i=1}^r [-N_i, N_i] \cap \mathbb{Z}^r, \quad \phi = \phi_Q|_W, \quad \lambda'(n_1, \dots, n_r) = \sum_{i=1}^r n_i \lambda(q_i),$$

where ϕ_Q is the coefficient identification of the proper GAP Q ; then $|B'| = |Q|$ and ϕ is injective. Lemma 3.3 gives $\lambda'(\phi(v)) = \lambda(v)$ for $v \in W$. The preceding bounds for $|Q|$ give alternatives (i) and (ii), and the identity of functional values proves the final assertion. $\square$

7. THE DENSITY-INCREMENT ITERATION

We use the exponent sequence defined in Section 2.

Lemma 7.1 (Uniform up-move numerology). *Fix $D \geq 2$ and $\zeta_0 > 0$. There are $\varepsilon_0, c_* > 0$ such that, whenever*

$$1 \leq d < e \leq D, \quad \zeta_0 \leq \zeta \leq 1 - \sigma_d, \quad 0 < \varepsilon < \varepsilon_0,$$

one has

$$(\sigma_e + \zeta) \left(\frac{1}{\sigma_d + \zeta} - (1 - \varepsilon)(e - d) \right) \leq 1 - c_*$$

whenever the parenthetical expression is nonnegative.

Proof. For $2 \leq d < e$ one has $\sigma_d = \alpha_{d-1}$ and $\sigma_e = \alpha_{e-1}$ in the notation of [5], so the pointwise inequality, with a gap depending continuously on ζ , is [5, Observation 15] with all dimensions shifted down by one. Only the pairs with $d = 1$ are new. For $k = e - 1$, set

$$f_{1,k}(\zeta) = (\sigma_{1+k} + \zeta) \left(\frac{1}{\sigma_1 + \zeta} - k \right).$$

Direct substitution gives $f_{1,k}(0) = \sigma_{1+k}(5 - k) \leq 1$, with equality only for $e \in \{2, 3, 4\}$, and

$$f'_{1,k}(\zeta) = \frac{\sigma_1 - \sigma_{1+k}}{(\sigma_1 + \zeta)^2} - k < 0,$$

so the gap is strictly positive at $\varepsilon = 0$ for every $\zeta \in [\zeta_0, 1 - \sigma_1]$ for which the parenthetical factor is nonnegative. There are finitely many pairs $1 \leq d < e \leq D$; compactness and continuity give a uniform gap $c_* > 0$ over the stated ranges, which persists after choosing $\varepsilon_0 > 0$ sufficiently small. $\square$

Theorem 7.2 (Projective theorem). *For every fixed d and $\beta > 1$, every (λ, X) -shell non-dividing set $V \subseteq B \subseteq \mathbb{Z}^d$, where B is a symmetric coordinate box and $|B| \leq |V|^\beta$, satisfies*

$$|V| \leq |B|^{\sigma_d + o(1)}.$$

Here the $o(1)$ is uniform over the shell functional and shell parameter once d and β are fixed.

Proof. Assume for a contradiction that for some fixed $\zeta_0 > 0$ there are arbitrarily large examples

$$A_0 \subseteq B_0 \subseteq \mathbb{Z}^{d_0}, \quad |A_0| > |B_0|^{\sigma_{d_0} + \zeta_0}.$$

Write $n_0 = |A_0|$. Since $|A_0| \leq |B_0|$, the assumed examples are already impossible when $\sigma_{d_0} + \zeta_0 \geq 1$; hence we may and do assume $\sigma_{d_0} + \zeta_0 < 1$. All sets constructed below remain shell non-dividing, because the only operations are passage to subsets and coefficient identifications of proper homogeneous GAPs, with the shell functional transported by Lemma 3.3.

Choice of parameters. At any noncontradictory stage $A \subseteq B \subseteq \mathbb{Z}^d$ with excess at least ζ_0 , the inequalities $|A| \leq |B|$ and $|A| > |B|^{\sigma_d + \zeta_0}$ give

$$|B| < |A|^{\beta_*}, \quad \beta_* = (\sigma_1 + \zeta_0)^{-1} > 1.$$

Thus one fixed polynomial-box exponent works at every stage. Since $\sigma_r \rightarrow 1$, the integer

$$D_0 = \max\{r \geq 1 : \sigma_r \leq 1 - \zeta_0\}$$

is finite. At the beginning of every noncontradictory stage, the exact excess is at least ζ_0 and $|A| \leq |B|$, so its ambient dimension is at most D_0 .

For each $1 \leq \ell \leq D_0$, let G_ℓ be a bound for the output dimension in Lemma 3.4 with ambient dimension ℓ and polynomial-box exponent β_* . Choose

$$D > \max\{D_0, G_1, \dots, G_{D_0}\}.$$

Thus every one-step preprocessing output has dimension at most D , while every stage that survives to the next iteration has dimension at most D_0 . Apply Lemma 7.1 with D and ζ_0 , obtaining a uniform up-move gap $c_* > 0$.

Put

$$\Delta_D = \min_{2 \leq r \leq D} (\sigma_r - \sigma_{r-1}) > 0.$$

Choose $\varepsilon > 0$ so small that

$$\varepsilon < \min \left\{ \varepsilon_0, \frac{\zeta_0}{4}, \frac{c_*}{4}, \frac{\Delta_D}{8}, \frac{1}{3} \right\}$$

and so that $\kappa_{r-1} + \varepsilon < \sigma_r + \zeta_0$ for every $2 \leq r \leq D$. Choose K_0 sufficiently large that

$$\frac{1}{K_0} < \sigma_1 + \zeta_0, \quad K_0(\sigma_1 + \zeta_0) > 2.$$

For $1 \leq \ell \leq D_0$, let $K_\ell = K(\ell, \beta_*, \varepsilon, K_0)$ be an admissible exponent in Lemma 3.4, and put

$$K = \max_{1 \leq \ell \leq D_0} K_\ell.$$

Using this larger exponent is valid in every ambient dimension at most D_0 . By the final assertion of Lemma 3.4, there is a fixed number $\beta_{\text{pkg}} > 1$, depending only on the constants chosen so far, which bounds the polynomial-box exponent of every structural package output by preprocessing in dimensions at most D_0 . Put

$$\beta^\dagger = \max \left\{ \beta_{\text{pkg}}, \frac{2\beta_*}{1 - \varepsilon} \right\}.$$

Finally choose $0 < \kappa < 1$ sufficiently small in the fixed parameters.

We now choose a slowly varying scale. For each $1 \leq r \leq D - 1$, let $N_r(t)$ be a threshold for Lemma 6.1 in dimension r at parameter $\mu = t$. A diagonal choice gives a nondecreasing integer-valued function $\omega(n) \rightarrow \infty$ such that

$$\omega(n)^{2K+2} = o(\log n), \quad N_r(\omega(n)^{-\kappa}) \leq n^{1-\varepsilon} \quad (1 \leq r \leq D - 1);$$

explicitly, $\omega(n) = \max\{m : L_m \leq n\}$ for any integers $L_m \uparrow \infty$ with $\log L_m \geq m^{4K+4}$ and $L_m^{1-\varepsilon} \geq \max_r N_r(m^{-\kappa})$. No quantitative dependence of $N_r(t)$ on t is required.

At a stage with $n = |A|$, use

$$\delta = \omega(n)^{-1}, \quad \gamma = \delta^K, \quad \mu = \delta^\kappa.$$

Uniformly as $n \rightarrow \infty$, these parameters satisfy

$$\begin{aligned} \delta &\geq n^{-1/2}, & \gamma &\geq n^{-1/3}, & \delta &= o(\mu), \\ \frac{1}{\log n} &= o(\mu\gamma), & \gamma^{-1} &= o(\mu\gamma n), & [s(\delta n)]^{-1} &= o(\gamma). \end{aligned}$$

Each condition follows from $\omega(n) \rightarrow \infty$ together with the growth condition $\omega(n)^{2K+2} = o(\log n)$, which also forces $\omega(n) = n^{o(1)}$; we omit the routine verification, which is carried out in full in the formalization. The parameters therefore meet all hypotheses of the preprocessing theorem, and the last condition in the definition of ω will supply the largeness hypothesis in the weighted convex-density lemma after preprocessing.

The one-step increment. Suppose $A \subseteq B \subseteq \mathbb{Z}^d$ is a current stage, put $n = |A|$, and define its exact excess by

$$p = \frac{\log n}{\log |B|} = \sigma_d + \zeta.$$

The case $|B| = 1$ is already impossible for large n . Since $n \leq |B|$, one has

$$\zeta_0 \leq \zeta \leq 1 - \sigma_d.$$

In particular $\sigma_d \leq 1 - \zeta_0$, so $d \leq D_0$. Apply Lemma 3.4 with the ambient-dimension parameter d , and denote its output by $\tilde{A} \subseteq \tilde{B} \subseteq \mathbb{Z}^e$. Put $\tilde{n} = |\tilde{A}|$. Since $\tilde{n} > n^{1-\varepsilon}$ and $\log \tilde{n} \asymp \log n$, the same parameters satisfy, with $\tilde{n}$ in place of n ,

$$\delta \geq \tilde{n}^{-1/2}, \quad \frac{1}{\log \tilde{n}} = o(\mu\gamma), \quad \gamma^{-1} = o(\mu\gamma\tilde{n}), \quad [s(\delta\tilde{n})]^{-1} = o(\gamma).$$

In every preprocessing alternative one has $|\tilde{B}| \ll n^{\beta^*}$; since $\tilde{n} > n^{1-\varepsilon}$, it follows, after absorbing the fixed implicit constant, that

$$|\tilde{B}| \leq \tilde{n}^{\beta^\dagger}$$

for all sufficiently large n . By the definition of $\beta^\dagger$, the retained structural package arises with polynomial-box exponent at most $\beta^\dagger$. Thus the same package is used in every subsequent application of projective convexity, slab exclusion, filling and the index lemma. Finally,

$$\tilde{n} > n^{1-\varepsilon} \geq N_r(\mu) \quad (1 \leq r \leq D-1),$$

by the defining property of ω . Thus every subsequent use of projective convexity and weighted convex density is legitimate for the post-preprocessing set.

The dimension-changing cases are handled exactly as Cases 1 and 2 of the proof of [5, Theorem 2], with [5, Observation 15] replaced by Lemma 7.1. If $e > d$, put $a = \frac{1}{p} - (1-\varepsilon)(e-d)$; the box estimate gives $|\tilde{B}| \ll n^a$, which is impossible for large n when $a \leq 0$, and otherwise Lemma 7.1 gives $(\sigma_e + \zeta)a \leq 1 - c_*$, whence, after decreasing a constant $c_{\text{up}} = c_{\text{up}}(D, \zeta_0) > 0$ and absorbing the fixed implicit constant,

$$|\tilde{A}| > n^{1-\varepsilon} > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{up}}}.$$

Any noncontradictory output satisfies $\sigma_e + \zeta + c_{\text{up}} < 1$, hence $e \leq D_0$, so the stage-dimension invariant is preserved. If $e < d$, then $|\tilde{A}| > n^{1-\varepsilon}$ and $|\tilde{B}| \ll |B|$; since $p \leq 1$ and $\sigma_d - \sigma_e \geq \Delta_D$, the choices above give, after absorbing the fixed constant,

$$|\tilde{A}| > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{down}}}$$

for a fixed $c_{\text{down}} > 0$ depending only on D and ζ_0 .

It remains to consider $e = d$. Write

$$|\tilde{A}| = \rho n, \quad |\tilde{B}| \ll \rho^{K_0} |B|.$$

The pair $(\tilde{A}, \tilde{B})$ is irreducible. Let $\tilde{\lambda}$ and X be the transported shell functional and shell parameter. If $d = 1$, Theorem 5.4 says that its projective indexed family is in μ -convex position. It lies in the zero-dimensional affine space $H_{\tilde{\lambda}}$, whose only nonempty half-space is the whole space; that half-space contains every index. Since $\mu < 1$, this is impossible.

Assume $d \geq 2$. By Lemma 4.6, the family $\pi(\tilde{A})$ affinely spans $H_{\tilde{\lambda}}$ and lies in the convex body

$$\Omega = \{y \in H_{\tilde{\lambda}} : Xy \in \tilde{B}_{\mathbb{R}}\},$$

which has nonempty relative interior. Theorem 5.4 and Lemma 6.1, applied in the Euclidean affine space $H_{\tilde{\lambda}} \cong \mathbb{R}^{d-1}$, followed by Lemma 6.2, give a shell non-dividing set A' with

$$|A'| \geq \rho \eta^{a_d} n, \quad a_d = \kappa_{d-1} + \varepsilon, \quad \eta \in [\mu, \mu^{\hat{\tau}_d}],$$

where

$$\hat{\tau}_d := \tau_{d-1} > 0$$

depends only on d and ε . If cap lifting lowers the dimension to some $f < d$, then its box B' satisfies $|B'| \ll |\tilde{B}| \ll |B|$. Moreover $\eta^{a_d} = n^{-o(1)}$, and hence for sufficiently large n ,

$$|A'| \geq n^{1-2\varepsilon}.$$

The same calculation as in the preceding down case gives

$$|A'| > |B'|^{\sigma_f + \zeta + c_{\text{down}}}.$$

Thus every dimension-lowering cap also produces a fixed excess increment.

Suppose instead that cap lifting preserves dimension d . Then

$$|B'| \ll \eta \rho^{K_0} |B|.$$

At the old exponent $p = \sigma_d + \zeta$ we have

$$\frac{|A'|}{|B'|^p} \gg \eta^{a_d - p} \rho^{1 - K_0 p}.$$

The choices of ε and K_0 give, uniformly for $2 \leq d \leq D$,

$$p - a_d \geq \nu_0 > 0, \quad K_0 p - 1 \geq 1.$$

Let

$$\tau_* = \min_{2 \leq r \leq D} \hat{\tau}_r > 0.$$

Since $\eta \leq \mu^{\hat{\tau}_d}$ and $\rho \leq 1$, the right-hand side is at least

$$C_D^{-1} \mu^{-\tau_* \nu_0},$$

which tends to infinity. Therefore, for all sufficiently large n ,

$$|A'| > |B'|^p.$$

The exact excess strictly increases in a same-dimensional step. In addition,

$$\frac{|B'|}{|B|} \ll_D \eta \rho^{K_0} \leq C_D \mu^{\tau_*} = o(1).$$

This completes the one-step analysis.

The iteration terminates. We first prove that the iteration cannot lose too many points. An upward step increases the excess by at least c_{up} , while the exact excess is always at most $1 - \sigma_d < 1$. Hence there are only $O_{D, \zeta_0}(1)$ upward steps. Every downward step decreases the integer dimension, and between two upward steps there can be at most D downward steps. Since all dimensions are at most D , the total number of dimension-changing steps is at most some fixed number $M = M(D, \zeta_0)$.

For $2 \leq r \leq D$ put

$$c_r = \max \left\{ \kappa_{r-1} + \varepsilon, \frac{1}{K_0} \right\}.$$

By construction

$$c_r < \sigma_r + \zeta_0.$$

Consider any consecutive block of same-dimensional steps in dimension r . At a step in this block, let ρ_j be the preprocessing cardinality ratio and let η_j be the cap-volume ratio. Put

$$x_j = \log(1/\eta_j), \quad y_j = \log(1/\rho_j).$$

The point and box estimates give

$$\begin{aligned} \log \frac{|A_j|}{|A_{j+1}|} &\leq (\kappa_{r-1} + \varepsilon) x_j + y_j, \\ \log \frac{|B_j|}{|B_{j+1}|} &\geq x_j + K_0 y_j - O_D(1). \end{aligned}$$

The first right-hand side is at most $c_r(x_j + K_0 y_j)$. Also

$$x_j \geq \tau_* \log(1/\mu_j) = \tau_* \kappa \log \omega(|A_j|) \longrightarrow \infty$$

uniformly whenever all current cardinalities are bounded below by a fixed positive power of n_0 , because ω is nondecreasing and tends to infinity. Thus the $O_D(1)$ term is $o(x_j + K_0 y_j)$ uniformly. Summing over any initial segment of the block gives

$$\log \frac{|A_{\text{start}}|}{|A_{\text{end}}|} \leq (c_r + o(1)) \log |B_{\text{start}}|.$$

Since the starting excess is at least ζ_0 ,

$$|A_{\text{end}}| \geq |A_{\text{start}}|^{1-c_r/(\sigma_r+\zeta_0)-o(1)}.$$

Therefore there is a fixed $\theta > 0$, depending only on the chosen parameters, such that every same-dimensional block, including every partial block, retains at least the θ -th power of its input cardinality. Every dimension-changing step retains at least $|A_j|^{1-2\varepsilon}$ points.

There are at most M dimension-changing steps and at most $M+1$ same-dimensional blocks. Hence there is a fixed $q > 0$ such that every iterate produced before a contradiction satisfies

$$|A_j| \geq n_0^q.$$

To remove the apparent circularity in the uniform $o(1)$ above, run the construction with the provisional stopping rule $|A_j| < n_0^{q/2}$. On the range before this rule triggers, all $o(1)$ estimates are uniform; the preceding product calculation yields the stronger lower bound $|A_j| \geq n_0^q$, so the provisional rule never triggers.

We may therefore iterate indefinitely unless a contradiction occurs. There are only finitely many dimension changes. After the last one, every step has the same dimension, and the box estimate above gives, uniformly for all remaining stages and sufficiently large n_0 ,

$$|B_{j+1}| \leq \frac{1}{2}|B_j|.$$

A positive integer box cardinality cannot be halved indefinitely, while the accompanying sets all have cardinality at least $n_0^q > 1$. This is the required contradiction. $\square$

8. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.2. The lower bound $F(N) \gg N^{1/5}$ is implicit in the constructions of Erdős and of Erdős–Lev–Rauzy–Sándor–Sárközy [3, 4] (see also [1]): translating a Bosznay-type non-averaging set of size m and diameter D by any $x > mD - \min B$ turns every divisibility relation into an averaging relation, and optimizing the parameters places such a set in $[2m^5]$. For the upper bound, fix $0 < \varepsilon < 1$ and suppose, for arbitrarily large N , that there is a non-dividing set $A \subseteq [N]$ with

$$|A| > N^{1/5+2\varepsilon}.$$

Partition $[N]$ into dyadic intervals. Some interval $[X, 2X)$ contains a subset $A_0 \subseteq A$ with

$$|A_0| \geq \frac{|A|}{1 + \log_2 N} > N^{1/5+\varepsilon}$$

for all sufficiently large N . Take

$$d = 1, \quad B = [-2X, 2X] \cap \mathbb{Z}, \quad \lambda(t) = t.$$

Then A_0 is shell non-dividing and $|B| \leq 4N + 1$. Consequently, for all sufficiently large N ,

$$|A_0| > |B|^{1/5+\varepsilon/2}, \quad |B| < |A_0|^\beta, \quad \beta = (1/5 + \varepsilon/2)^{-1} > 1.$$

Theorem 7.2, applied with this fixed β , gives

$$|A_0| \leq |B|^{1/5+o(1)}.$$

For sufficiently large N the $o(1)$ term is smaller than $\varepsilon/2$, contradicting the preceding strict inequality. Hence $F(N) \leq N^{1/5+o(1)}$, and the theorem follows. $\square$

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